Paper-to-Peer (P2P)

_ Controllability of Multi-agent systems from a graph-theoretic perspective _

Oct. 26. 2026 (Week 6)

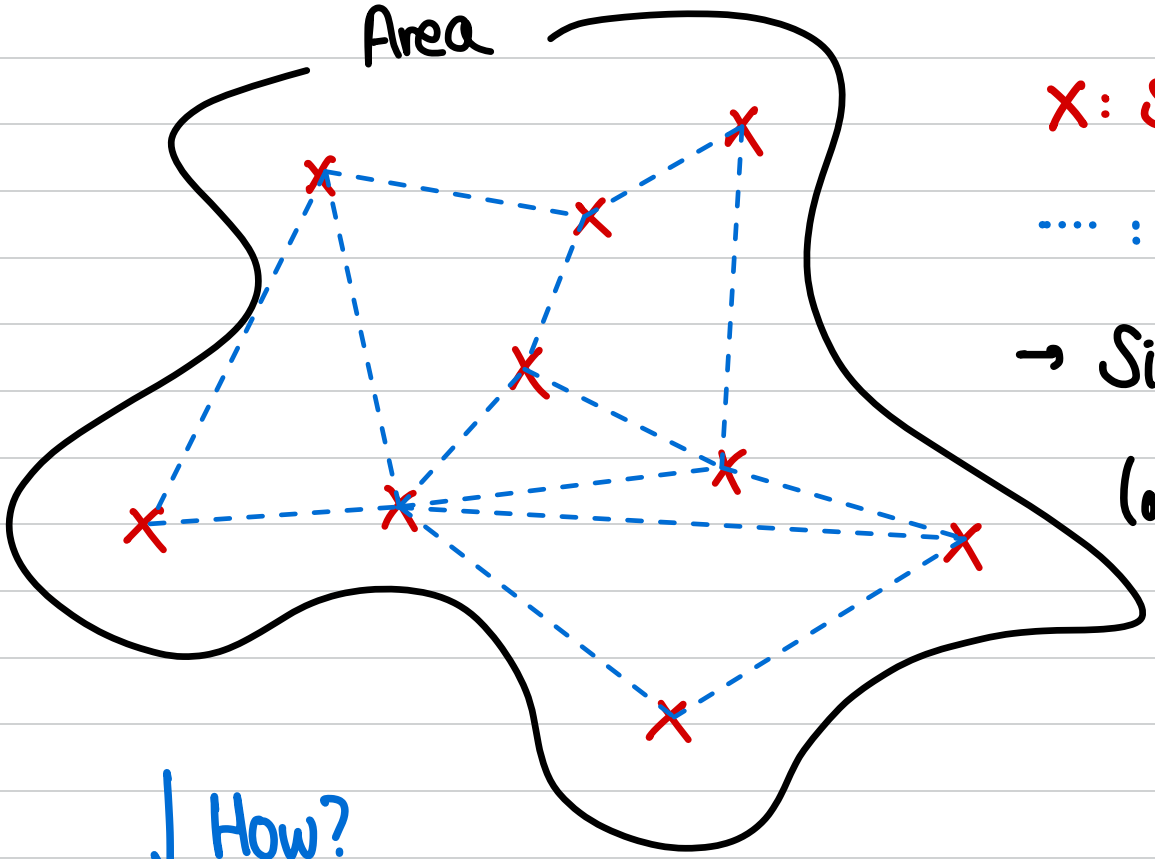
Woojin Shin

Department of Mathematical Science, GIST

* Review

Q. What was the major concern in prev. presentation?

↳ Agreement (Consensus) of the network.



X : Sensors

···· : Network

→ Single Representative?

(or every node behave identically?)

↓ How?

Every node follows the "Agreement Protocol" $\dot{x}(t) = -L(G)x(t)$

Explicitly, $\dot{x}_i(t) = -\sum_{j \sim i} (x_i(t) - x_j(t))$

(using local info., every node evolves to decrease the total relative state difference.)

Thm) If every node follows the agreement protocol,
G is connected iff $x(t) \to \bar{x}_0 \mathbb{1} \in \text{span}\{\mathbb{1}\}$.

In this chapter move our interest on to "Control."

Q. What is Control?

↳ Make the system to show the desired behavior.

Formally, ① given an initial state $x(0) = x_0$,

② by applying external control input $u(t)$,

③ Send the state to our desired state in finite time.

Dynamics : $\dot{x}(t) = f(x(t), u(t))$

$\forall x(0) = x_0$, find $u(t)$ s.t. $\exists T > 0 \cdot x(T) = x_d$.

* Preliminaries (Linear Systems Theory)

· Linear System ?

$$\begin{cases} \dot{x}(t) = A(t)x(t) + B(t)u(t) \\ y(t) = C(t)x(t) + \cancel{D(t)u(t)}\ 0 \end{cases}$$

$A \in \mathbb{R}^{n \times n}$: System matrix

$B \in \mathbb{R}^{n \times m}$: Input matrix

$C \in \mathbb{R}^{p \times n}$: Output matrix

If $A(t) = A$ and $B(t) = B$, then the system be

Linear Time - Invariant (LTI).

Def) Open loop / Closed loop System

The input $u(t)$ is { independent to the state $x(t)$ / a function of states ; $u(t) = f(x(t))$

then, the system is open / closed.

Def) Reachable State

A state $z \in \mathbb{R}^n$ is called reachable from $x(0) = x_0$ if

$\exists$ Closed loop control $u(t)$ s.t. $x(T) = z$ for finite $T > 0$.

Def) Controllability

The system is called controllable if any state $z \in \mathbb{R}^n$ is reachable from an arbitrary initial state $x(0) = x_0$.

Thm) Equivalent Conditions for Controllability (PBH test)

1) The LTI system (A, B) is controllable.
2) $\forall \lambda \in \mathbb{C}$, $\operatorname{rank}[A - \lambda I \mid B] = n$.
3) For all left eigenvectors $w \in \mathbb{C}^n$ of A, $B^T w \neq 0$.
   not orthogonal

Def) Observability

The system is called observable on $[t_0, t_1]$ if any initial state x_0 can be uniquely determined from the output $y([t_0, t_1])$ and the input $u([t_0, t_1])$.

Thm) Equivalent conditions for observability (PBH test)

1) The LTI system (A, C) is observable.
2) $\forall \lambda \in \mathbb{C}$, $\operatorname{rank}[(A-\lambda I)^T | C^T]^T = n$.
3) For all eigenvectors $w \in \mathbb{C}^n$ of A, $Cw \neq 0$.

* Intuition

Recall : Solution space of ODE

$\dot{x} = Ax \rightarrow x(t) = \sum_i a_i e^{\lambda_i t} v_i$. (Linear combi. of modes)

Here. $a_i = z_i(0)$ where $z_i(t) = w^T x_i(0)$.

For $\dot{x} = Ax + Bu$, and left eigenvector w of A,

$$w^T\dot{x} = w^TAx + w^TBu = \lambda w^Tx + (w^TB)u \Rightarrow \dot{z} = \lambda z + (w^TB)u$$

If $w^TB = 0$, then $\dot{z} = \lambda z + 0u$

→ Input cannot affect the mode corrsp. to λ!

§10.1. Basic Input - Output Setup

* Standing assumption: input node = output node

* Partition of nodes.

  - input nodes: influencing the network and observing neighbors.
  - floating nodes: just follows the ambient agreement protocol.

  • In some specific frameworks, they can be considered as

    Leaders & Followers

So, the incidence matrix $D(G)$ is partitioned by blocks

$$D(G) = \left(\begin{array}{c} D_f \\ \hline D_i \end{array}\right) \begin{array}{l} \} \text{ floating vertices } (D_f \in \mathbb{R}^{n_f \times m}) \\ \} \text{ input vertices } (D_i \in \mathbb{R}^{n_i \times m}) \end{array}$$

and since $L(G) = D(G)D(G)^T$,

$$L(G) = \begin{pmatrix} D_f \\ D_i \end{pmatrix} \begin{pmatrix} D_f^T & D_i^T \end{pmatrix} = \begin{pmatrix} D_f D_f^T & D_f D_i^T \\ D_i D_f^T & D_i D_i^T \end{pmatrix}$$

$$\| \text{ let}$$

$$\begin{pmatrix} A_f & B_f \\ B_f^T & A_i \end{pmatrix}$$

By substituting such partition to the consensus protocol,

$$\dot{x}(t) = -\mathcal{L}(G)x(t) \iff \begin{pmatrix} \dot{x}_f(t) \\ \dot{x}_i(t) \end{pmatrix} = -\begin{pmatrix} A_f & B_f \\ B_f^T & A_i \end{pmatrix}\begin{pmatrix} x_f(t) \\ x_i(t) \end{pmatrix}$$

and therefore, $\begin{cases} \dot{x}_f(t) = -A_f x_f(t) - B_f x_i(t) \\ \dot{x}_i(t) = -B_f x_f(t) - A_i x_i(t) \end{cases}$

Note 1) $x_i(t)$ is an input node. So, $x_i(t) = u(t)$.

Note 2) Our assumption : Input node = Output node

└ $x_i(t)$: effect on output. → Choose y = effected by state.

→ $\begin{cases} \dot{x}_f(t) = -A_f x_f(t) - B_f u(t) \\ y(t) = -B_f^T x_f(t) \end{cases}$ (LTI system) (*)

Prop) PBH-variant for input-output network

The LTI system (*) is controllable and observable

if and only if none of the eigenvec. of A_f are simultaneously orthogonal to all columns of B_f

↳ Directly follows from PBH test of Ctr'bility & Obs'bility.

For the system (*), Controllable pair = (A_f, B_f)

Observable pair = (A_f, B_f^T)

Recall) PBH test

(A.B) Ctr'ble. $\iff$ For all left eigenvectors $w \in \mathbb{C}^n$ of A, $B^T w \neq 0$.

(A.C) Obs'ble. $\iff$ For all eigenvectors $w \in \mathbb{C}^n$ of A, $Cw \neq 0$.

- One result follows from the prev. prop.

Prop) Dimension argument of Ctrl'bility & Obs'bility

If A_f has an eigenvalue with geometric multiplicity ≥ 2,
then the system is uncontrollable & unobservable.

proof) Let λ : eigenvalue of A_f with geometric multiplicity ≥ 2.
Then, $\exists$ corrsp. eigenvectors v_1 and v_2.

Case 1) At least one of v_1 or v_2 are orthogonal to B_f.
↳ Directly uncontrollable & unobservable by prev page.

Case 2) Neither v_1 nor v_2 are orthogonal to B_f.

Note that $v = v_1 + cv_2$ is also an eigenvec. of A_f corrsp to λ.

By choosing $c = -v_1^T B_f / v_2^T B_f$, $\quad v = v_1 - \dfrac{v_1^T B_f}{v_2^T B_f} v_2$

since $v_1^T B_f$, $B_f^T v_2$ are scalars, they are transpose-invariant.

and hence $B_f^T v = B_f^T v_1 - \dfrac{v_1^T B_f \, B_f^T v_2}{v_2^T B_f} = 0.$

Thus, v is an eigenvec. of A_f corrsp to λ that is orthogonal to B_f.

By prev. prop., uncontrollable & unobs'ble. ■

There is one more result derived from PBH-variant

Lem) Controllability test w/ Shared eigenvalue

Given a connected network, (*) is controllable

if and only if $\mathcal{L}(G)$ and A_f do not share an eigenvalue

Proof) ($\Leftarrow$, contrap.) Recall: $\mathcal{L}(G)$ be partitioned by $\mathcal{L}(G) = \begin{pmatrix} A_f & B_f \\ B_f^T & A_i \end{pmatrix}$

Suppose that (*) is uncontrollable.

Then, $\exists v_i \in \mathbb{R}^{n_f}$ s.t. $A_f v_i = \lambda v_i$ for some λ & $B_f^T v_i = 0$.

$$\mathcal{L}(G)\, v = \begin{pmatrix} A_f & B_f \\ B_f^T & A_i \end{pmatrix} \begin{pmatrix} v_i \\ 0 \end{pmatrix} = \begin{pmatrix} A_f v_i \\ B_f^T v_i \end{pmatrix} = \begin{pmatrix} \lambda v_i \\ 0 \end{pmatrix} = \lambda v$$

($\Rightarrow$, contrap.) Let $A_f = P_f^T \mathcal{L}(G) P_f$ where $P_f = [I_{n_f}, 0]^T$ is $n_f \times n$

If A_f and $\mathcal{L}(G)$ share a common eigval. λ,

for corresp. eigvecs. v_f and v, $v = P_f v_f$.

Then, $\mathcal{L}(G)v = \begin{pmatrix} A_f & B_f \\ B_f^T & A_i \end{pmatrix} \begin{pmatrix} v_f \\ 0 \end{pmatrix} = \lambda \begin{pmatrix} v_f \\ 0 \end{pmatrix} \Rightarrow B_f^T v_f = 0$.

$\therefore$ Uncontrollable. ■

§ 10.2. Graph-Theoretic Controllability - SISO

Controllability & Observability of

SISO agreement protocol



relation

Underlying Network Structure

First, introduce some definitions related to graph structure.

Def) Permutation matrix

A permutation matrix is a binary matrix with only one nonzero(1) entry in each row and column.

ex) $$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix} = \begin{pmatrix} y \\ x \\ z \\ w \end{pmatrix}$$

Def) Input Symmetry

The system is input symmetric if there exists a non-identity permutation matrix J s.t.

$$JA_f = A_f J$$

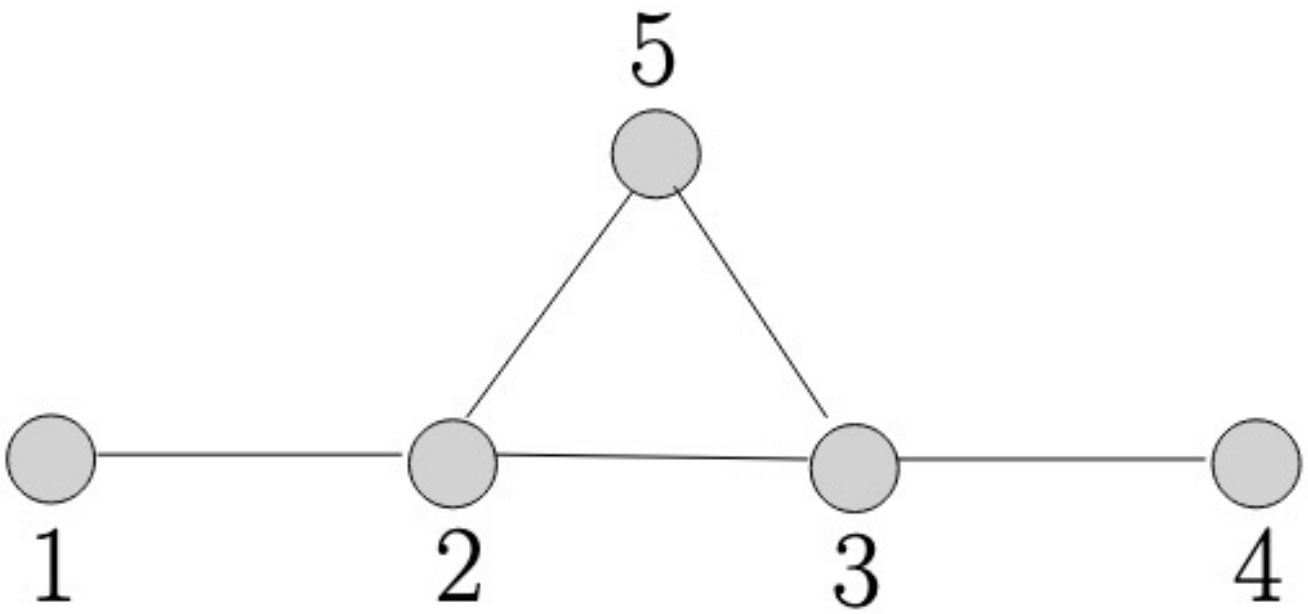
Meaning) Note that A_f governs the dynamics and J shuffles every node. So, $A_f J = JA_f$ implies that

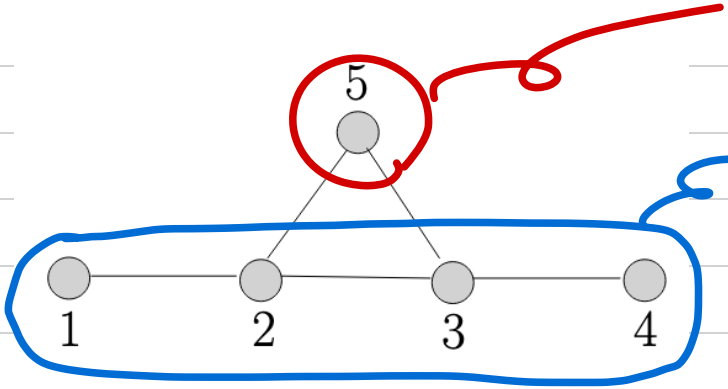
relabeling nodes after applying dynamics
=
relabling nodes before applying dynamics

→ Symmetry structure in graph(network)

Quiz.

Is there any node that makes this graph input symmetric?





Choose 5 as an input node.

Then, the others 1. 2. 3. and 4 be floating vertices.

- Graph Laplacian $L(G) =$ $\left(\begin{array}{cccc|c} 1 & -1 & 0 & 0 & 0 \\ -1 & 3 & -1 & 0 & -1 \\ 0 & -1 & 3 & -1 & -1 \\ 0 & 0 & -1 & 1 & 0 \\ \hline 0 & -1 & -1 & 0 & 2 \end{array}\right)$ (A_f)

By letting $J = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}$, $JA_f = A_f J$

So, this graph is input symmetric w.r.t. node 5.

With the def. of input symmetry, we can introduce the main theorem of this section.

☆ Thm) Controllability test via Graph Structure (SISO)

The system $\begin{cases} \dot{x}_f(t) = -A_f x_f(t) - B_f u(t) \\ y(t) = -B_f x_f(t) \end{cases}$ is uncontrollable if it is input symmetric. The converse is not true.

Intuition) If the graph is input symmetric, Then, we cannot make the system state asymmetric.

(The influence of input flows symmetrically over the graph)

Proof) Consider the input-symmetric system, i.e.

$\exists$ permutation matrix J s.t. $JA_f = A_f J$.

If $\exists$ eigval. with geom. mult ≥ 2, Uncontrollable by the prev. prop.

Let all the eigenvalues of A_f have geom. mult 1.

Given an eigval.-eigvec. pair (λ, v), $JA_f v = J(\lambda v)$

By input-symmetry. $A_f(Jv) = \lambda(Jv)$ → Jv : eigvec.

Since the eigenspace E_λ has dim 1 for each eigval λ,

$Jv \in \text{span}(\{v\})$ and hence $v - Jv \in \text{span}(\{v\})$ as well.

→ $v - Jv$ is eigvec. of A_f corresp. to λ.

Moreover, from $B_f = -A_f \mathbb{1}$,

$$JB_f = -JA_f\mathbb{1} = -A_f(J\mathbb{1}) = -A_f\mathbb{1} = B_f$$

$$\rightarrow \underbrace{J^TJ}_{\| \; I} B_f = J^TB_f \qquad \therefore \; JB_f = J^TB_f = B_f.$$

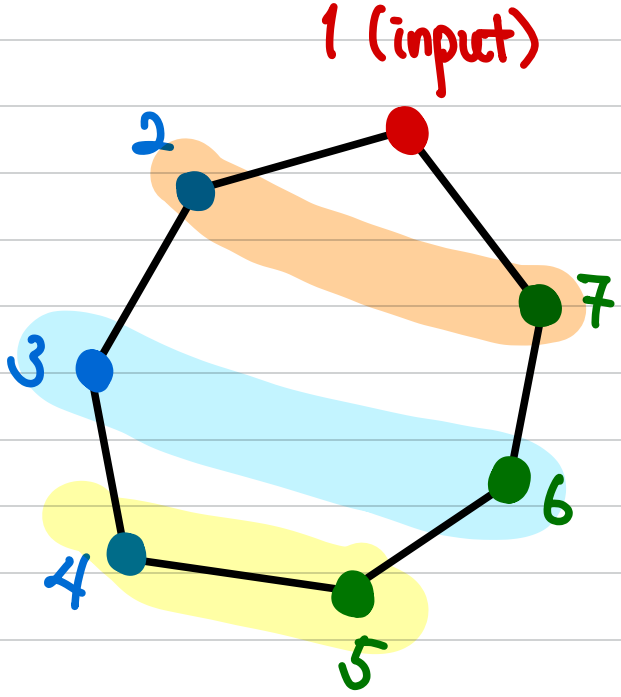
Therefore, $(\nu - J\nu)^TB_f = \nu^TB_f - (J\nu)^TB_f = \nu^TB_f - \nu^TJ^TB_f$

$$= \nu^TB_f - \nu^TB_f = 0$$

By the prev. prop., uncontrollable. ∎

Example

① Ring graph with SISO is uncontrollable.



Command

make $x = (1\ 2\ 3\ 4\ 5\ 6\ 7)^T$ by applying proper input.

But, the input signal applied at 1 flows symmetrically and hence

$x(2) = x(7)$, $x(3) = x(6)$, $x(4) = x(5)$ always need to be satisfied.

↳ Command cannot be completed → Uncontrollable!

- Analogous result for observability.

Thm) Observability Test

The system is unobservable if it is output - Symmetric.

Note: Duality between Controllability & Observability

(A, B) controllable $\iff$ (A^T, B^T) observable

§10.3. Graph-Theoretic Controllability - MIMO

- So far, we observed (un) controllability criterion for SISO case.
  - ↳ Q. How can we extend it to MIMO?

- One additional graph-theoretic tool: Equitable Partition

Def) Equitable Partition

Let π be r-partition of $V(G)$ with cells $C_1, C_2, \cdots, C_r$.

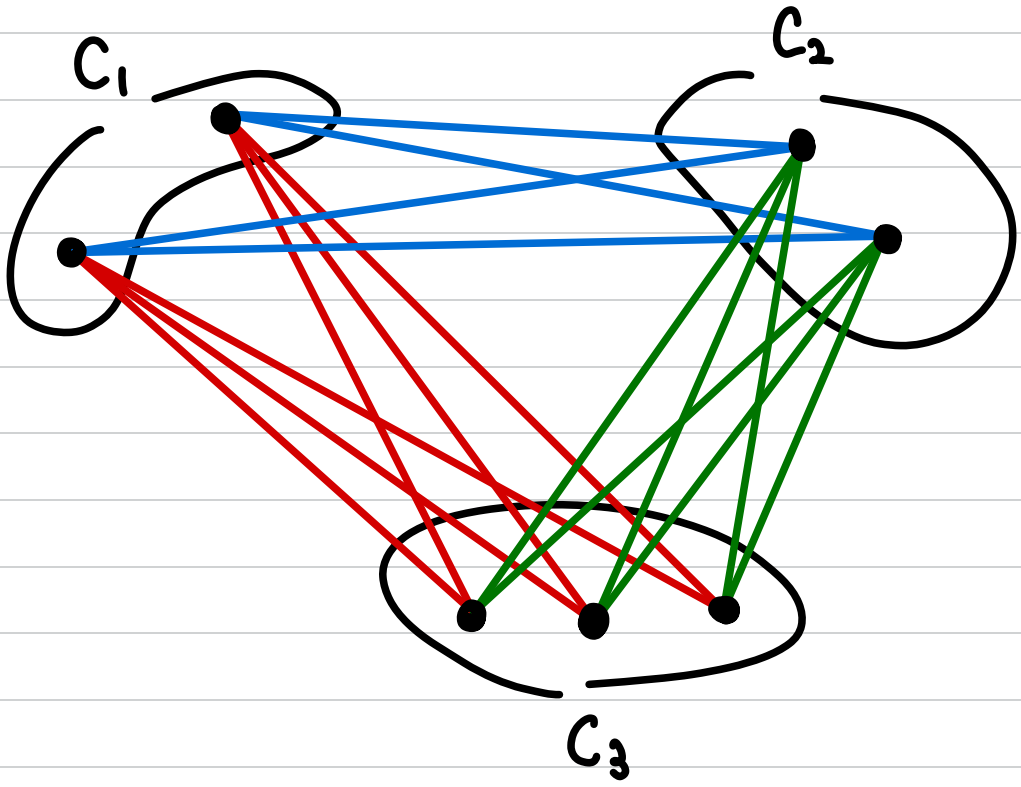
π is said to be equitable if

"for any pair i, j, each node in C_j has the same degree of neighbors in C_i."

Such degree is denoted by b_{ij}.

Example.

- Complete multipartite graph



$b_{11} = b_{22} = b_{33} = 0$

$b_{12} = 2$, $b_{13} = 2$

$b_{23} = 2$, $b_{21} = 2$

$b_{31} = 3$, $b_{32} = 3$

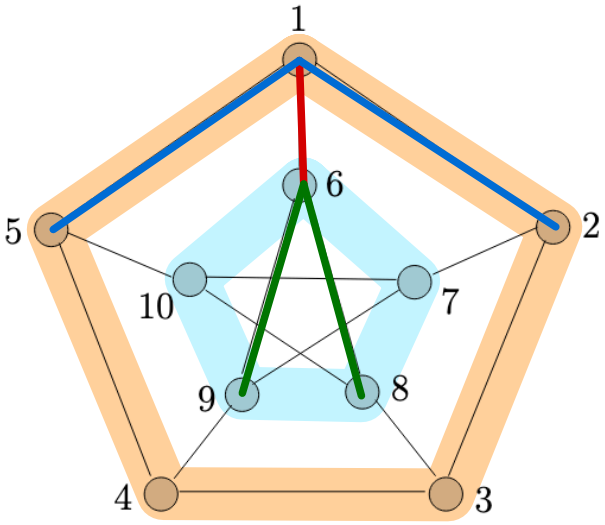
Def) Quotient Graph

Given a graph G and equitable r-partition π,

a quotient graph G/π is defined by a graph s.t.

$$[A(G/\pi)]_{ij} = b_{ij}$$

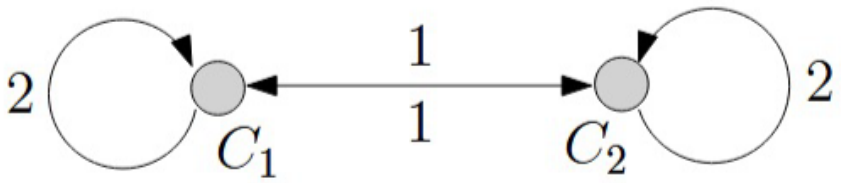
That is, a multigraph whose vertex is obtained by identifying each cell in equitable partition and the edges has weight b_{ij}.



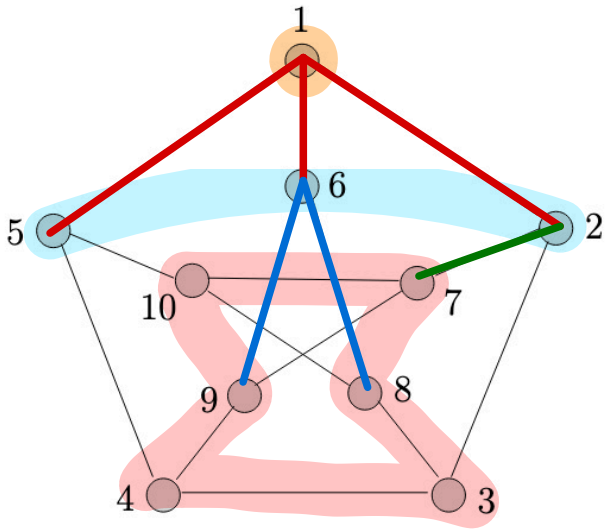
i) $\pi_1 = \{C_1, C_2\}$ where

$C_1 = \{1, 2, 3, 4, 5\}$, 2 1

$C_2 = \{6, 7, 8, 9, 10\}$ 2



$$A(\mathcal{G}/\pi_1) = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$



ii) $\pi_2 = \{C_1, C_2, C_3\}$ where

$C_1 = \{1\}$.

$C_2 = \{2, 5, 6\}$

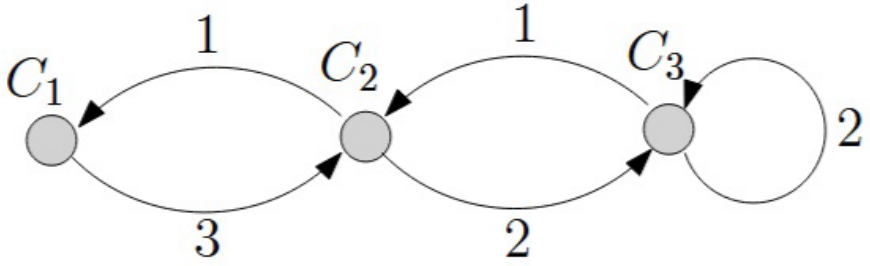
$C_3 = \{3, 4, 7, 8, 9, 10\}$

1

3

1

2



$$A(\mathcal{G}/\pi_2) = \begin{bmatrix} 0 & 3 & 0 \\ 1 & 0 & 2 \\ 0 & 1 & 2 \end{bmatrix}$$

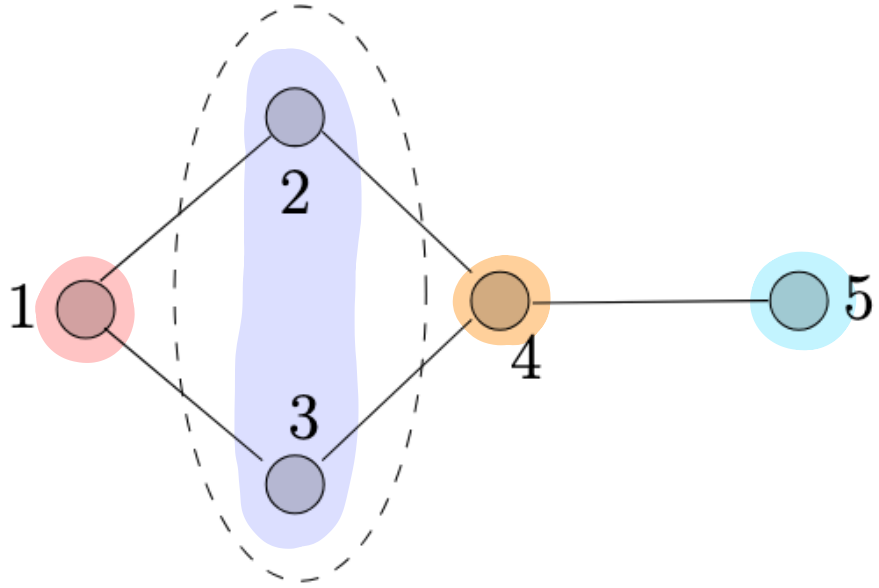
Def) Characteristic vector, Characteristic matrix

A characteristic vector $p_i \in \mathbb{R}^n$ of a nontrivial cell C_i has

1 in its components associated with C_i and 0 otherwise.

A characteristic matrix $P \in \mathbb{R}^{n \times r}$ of a partition π of V is a matrix

with Char. vecs. of the cells as its columns.



$$P = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Thm) Controllability test via Graph Structure (MIMO)

Given a connected graph G and the induced floating graph G_f.

the system $\begin{cases} \dot{x}_f(t) = -A_f x_f(t) - B_f u(t) \\ y(t) = -B_f x_f(t) \end{cases}$ is uncontrollable

if there exist Nontrivial Equitable Partitions on G and G_f.

Say π and π_f, such that all nontrivial cells of π are contained in π_f, that is, for all $C_i \in \pi \setminus \pi_f$, one has $|C_i| = 1$.

* Intuition

$$\dot{x}_f(t) = -A_f x_f(t) - B_f u(t)$$

↳ Solve: $x_f(t) = e^{-A_f t} x_f(0) - \int_0^t e^{-A_f(t-s)} B_f u(s)\, ds$

Matrix exponential: $e^{-A_f \tau} = \sum_{k=0}^{\infty} \frac{(-\tau)^k}{k!} A_f^k$

$$x_f(t) \in \text{span}\{B_f, A_f B_f, A_f^2 B_f, \cdots\}$$

$$= \text{span}\{B_f, A_f B_f, A_f^2 B_f, \cdots, A_f^k B_f\}$$

↓ Cayley-Hamilton

B_f: input matrix

$A_f^k B_f$: effect of input after k-hop

- π is equitable.

  ↳ "Cell" of π = Set of nodes having same connected property.

- Every nontrivial cell belongs to $\pi \setminus \pi_f$

  = Input (Leader) nodes are singleton cells.

  ⇒ Since input nodes are can always be controlled individually,

  they are distinguishable.

  & Follower nodes in same cells are not distinguishable.

→ So, the existence of NEP makes the system Uncontrollable.

Thank you.

It's the end of the presentation